



RAMCO INSTITUTE OF TECHNOLOGY (An Autonomous Institution)

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Department of Information Technology Academic Year 2025 - 2026 (Even Semester)

Semester, Degree & Branch: VI Semester, B.Tech. IT

Course Code & Title: CCS354 & Network Security

Name of the Faculty member (s): Mrs. G.Sivasathiya, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit IV / Mobile Device Security
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO12
- **Activity Chosen:** Flipped Classroom
- **Justification:**

A flipped classroom is an instructional strategy where students learn basic concepts before attending the class through videos, presentations, or study materials, while classroom time is used for discussions, problem-solving, and collaborative learning activities. This method improves student engagement, encourages active participation, enhances critical thinking, and promotes self-directed learning. It also allows students to understand mobile device security concepts at their own pace before applying them in classroom activities.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - The topic “Mobile Device Security” was selected for the flipped classroom activity. Students were instructed to learn the concepts independently and prepare for classroom discussion.
 - During the classroom session, S.Jeyadevi actively participated in discussions, answered questions, and clarified doubts related to mobile threats, authentication methods, secure applications, and device protection techniques
 - This method created an interactive learning environment where students became active participants instead of passive listeners. Fig 1.a shows the presentation done by S.Jeyadevi.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	2	2	3	2	3	3	3	2	1	2	1	3

(1 – Low

2 – Moderate

3 – High)

PO/PSO Mapped:

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	2	2	3	2	3	3	3	2	1	2	1	3
Justification for Correlation	Demonstrating network security practices involves applying basic engineering principles to understand security concepts and methodologies.	Carrying out advanced network security measures requires analyzing complex threats and identifying suitable security techniques to address them effectively.	Demonstrating network security practices involves designing and developing security solutions that cater to the specific needs of the system, contributing to effective protection and security outcomes	Advanced network security practices involve investigating complex threats and applying advanced security techniques to uncover underlying vulnerabilities and risks	Demonstrating network security practices includes utilizing modern security tools and technologies to automate security processes and enhance protection efficiency.	Effective network security practices require collaboration within teams and individual contributions to plan, implement, and analyze security measures	Effective communication of network security practices and results to stakeholders ensures alignment and support for security efforts, contributing to successful project outcomes	Demonstrating network security practices involves considering project timelines, budgets, and resource allocation to plan and implement security measures effectively.	Demonstrating network security practices requires a commitment to continuous learning to stay updated with evolving security threats and technologies	Demonstrating network security practices and carrying out advanced security measures align with a comprehensive understanding of network systems, ensuring the skills to identify and provide solutions for complex security challenges.	Advanced network security practices demonstrate the ability to develop skills in emerging technologies such as mobile application security, ensuring the delivery of reliable and secure IT solutions	Demonstrating advanced network security practices indicates strategies in securing systems using open-source tools to deliver reliable and secure solutions for business and industry needs

- **Images / Screenshot of the practice:**



Figure 1. a Mobile Device Security topic explained by S.Jeyadevi

Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

- Students appreciated learning the concepts before class as it helped them participate confidently in discussions.
- Students found the classroom sessions more interactive and engaging compared to traditional lecture methods.
- Faculty members observed improved student participation, collaboration, and critical thinking during the activity.

❖ ***Benefit of the practice:***

- Students actively participated in discussions and problem-solving activities.
- Pre-class preparation improved students' understanding and knowledge retention. The activity promoted collaborative learning, communication, and peer interaction.
- Students developed self-learning habits and became more responsible for their learning process.

❖ ***Challenges faced in implementation:***

- Managing classroom discussions within the allotted time was challenging.
- A few students initially required guidance to adapt to the flipped classroom learning approach

❖ **References:**

- <https://www.teachthought.com/learning/the-definition-of-the-flipped-classroom/>
- <https://www.panopto.com/blog/what-is-a-flipped-classroom/>

Signature of Faculty Member

HOD